

All the relevant code for the problems solved as part of this assignment can be found here:

https://colab.research.google.com/drive/1ScclrOluSVBqpEFZ34ZkTN_hswnkOUvO?usp=sharing

The Google Drive folder containing all relevant data, code and the report can be found here:

<https://drive.google.com/drive/folders/1VRxeAcLcg89Pcxi8fB137ZjKGqqXEYWY?usp=sharing>

1.

- The data comprising predictor variables was standardised for the purposes of unsupervised learning. The values associated with the target variable for different samples were not standardised since they were categorical values, which were used for labelling data points, thereby comparing and evaluating different clustering approaches.
- Agglomerative Clustering was performed using Ward Linkage, by using the default 'Euclidean Distance' as the distance metric. The 'maxclust' criterion was used to ensure that no more than $t = 3$ clusters were formed, while generating the clustering heatmap.
- The Dendrogram obtained is given as follows:

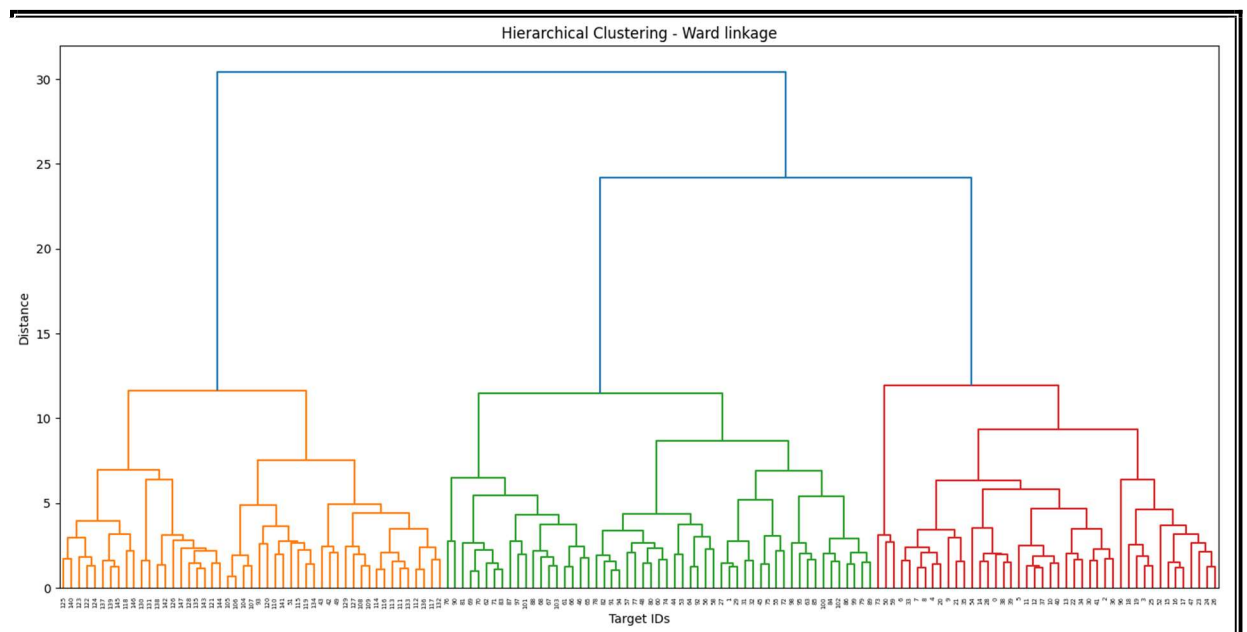


Figure 1: Dendrogram obtained on performing Hierarchical Clustering using Ward Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters

- The ARI and AMI values obtained are given as follows:

ARI	0.6807886831798285
AMI	0.6663995382342933

Table 1: ARI and AMI values for Hierarchical Clustering using Ward Linkage

- The positive value of ARI indicates that the accuracy of the clustering algorithm performed is better than that obtained by random chance. Since ARI is not equal to 1, we conclude that the results of clustering are not fully accurate.
- This may be the result of a data point, for instance, being present in different clusters in real and predicted ones, or certain points not present in a real cluster may have been grouped with the points that belong to the said cluster in the predicted cluster with the same label [1].
- For example, the sample with sample ID 54, which belongs to Class 1, has been incorrectly predicted to belong to Class 0.
- The corresponding heatmap with dendrogram is given as follows:

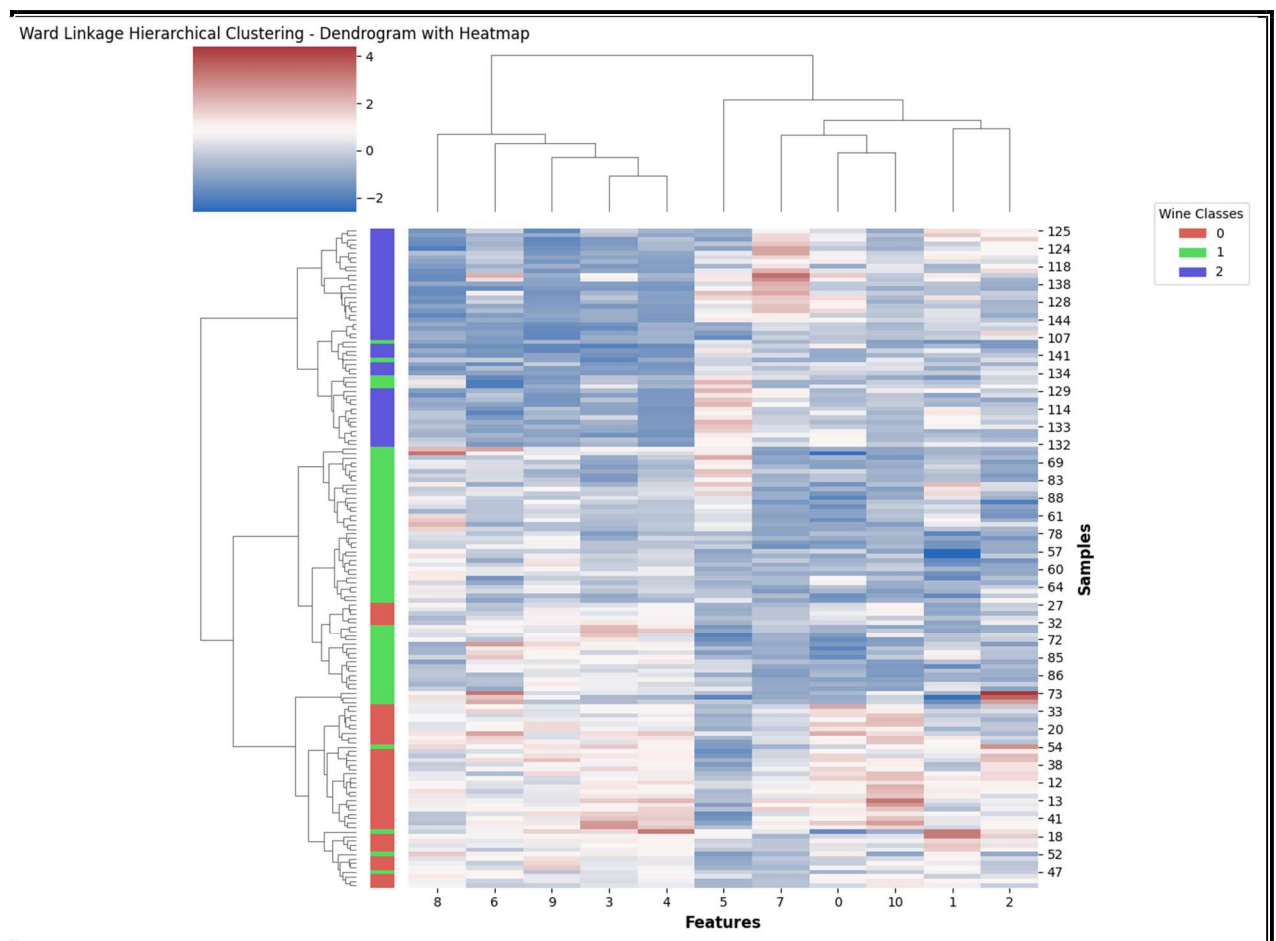


Figure 2: Heatmap with Dendrogram obtained on performing using Ward Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters

- We notice that several samples belonging to Class 1 have been clustered along with samples belonging to either Class 0 or Class 2, indicated by the relatively short green-coloured (Class 1) bands among wider red- (Class 0) and blue-coloured (Class 2) bands.

- Moreover, some samples belonging to Class 0 have been clustered along with samples belonging to Class 1, indicated by the relatively short red-coloured (Class 0) band flanked by wider green-coloured (Class 1) bands on either side.
- The AMI value indicates the extent of agreement between the labels of predicted and real clusters [1]. Here, $0 < \text{AMI} < 1$, thus indicating that while the predicted and real clusters with are similar to each other, but not identical; this is due to some erroneously clustered samples, an example of which is given above.

2.

- The Dendrograms obtained by performing Hierarchical Clustering using Single and Complete Linkage, and using 'Euclidean Distance' and 'Manhattan Distance' as the distance metrics are given as follows:

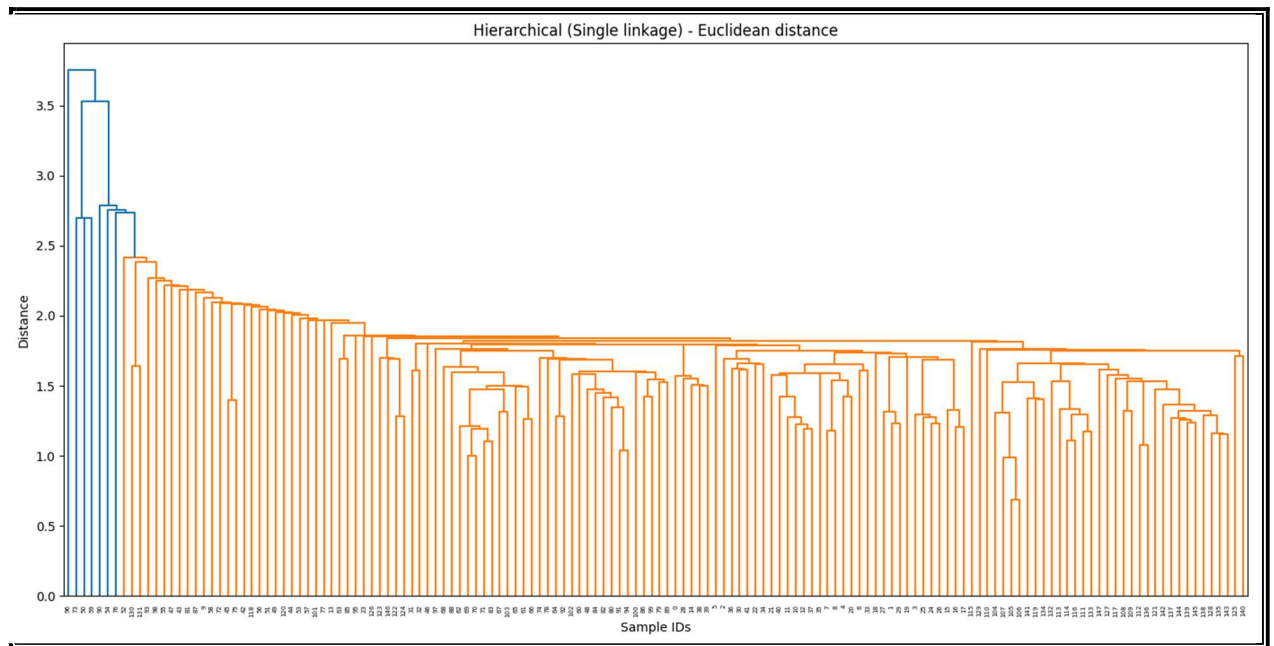


Figure 3: Dendrogram obtained on performing Hierarchical Clustering using Single Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters

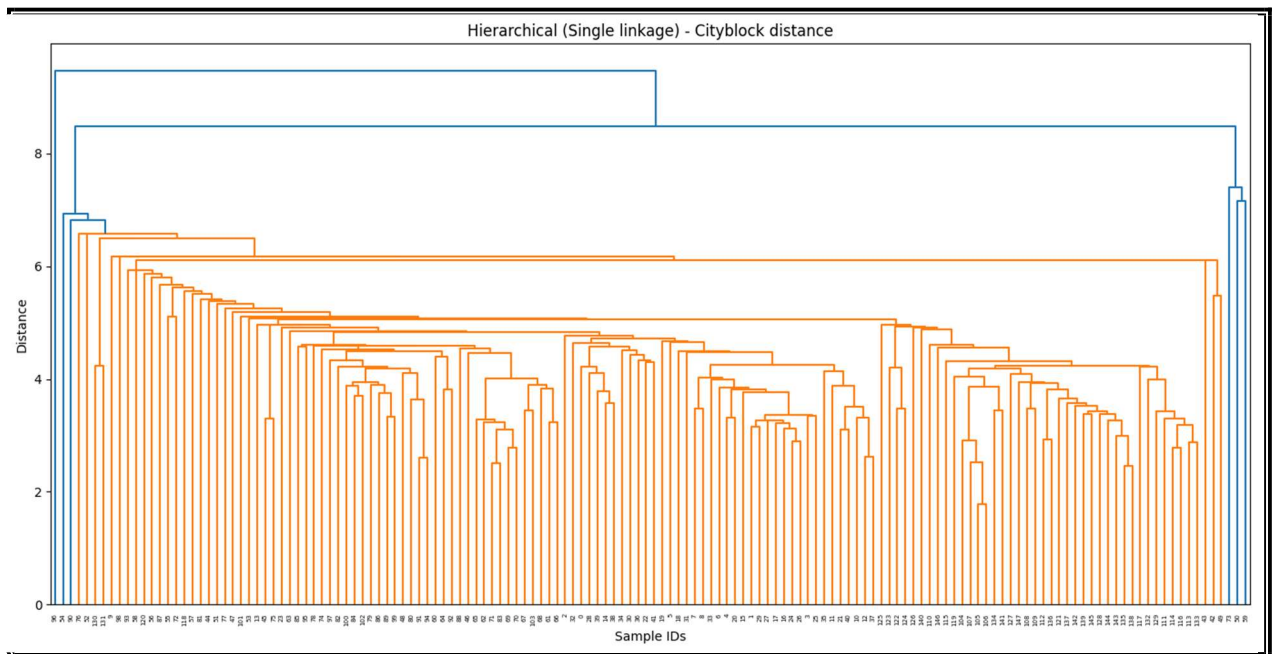


Figure 4: Dendrogram obtained on performing Hierarchical Clustering using Single Linkage with Manhattan Distance as the distance metric and $t = 3$ maximum number of clusters

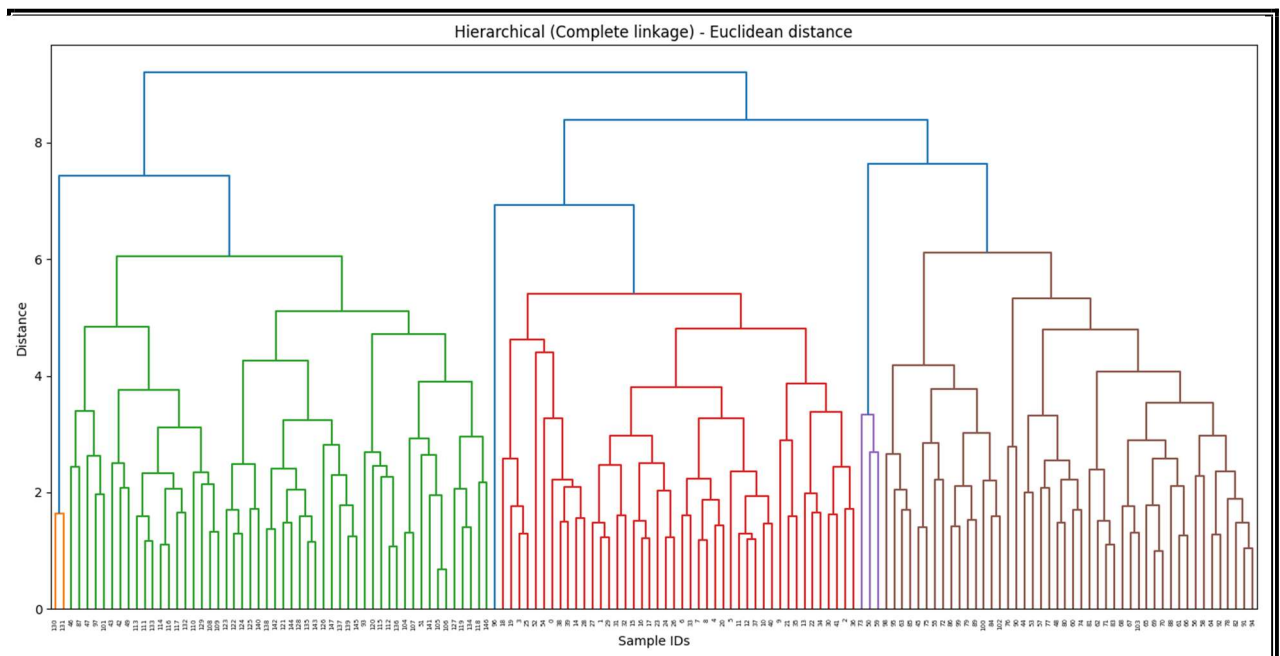


Figure 5: Dendrogram obtained on performing Hierarchical Clustering using Complete Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters

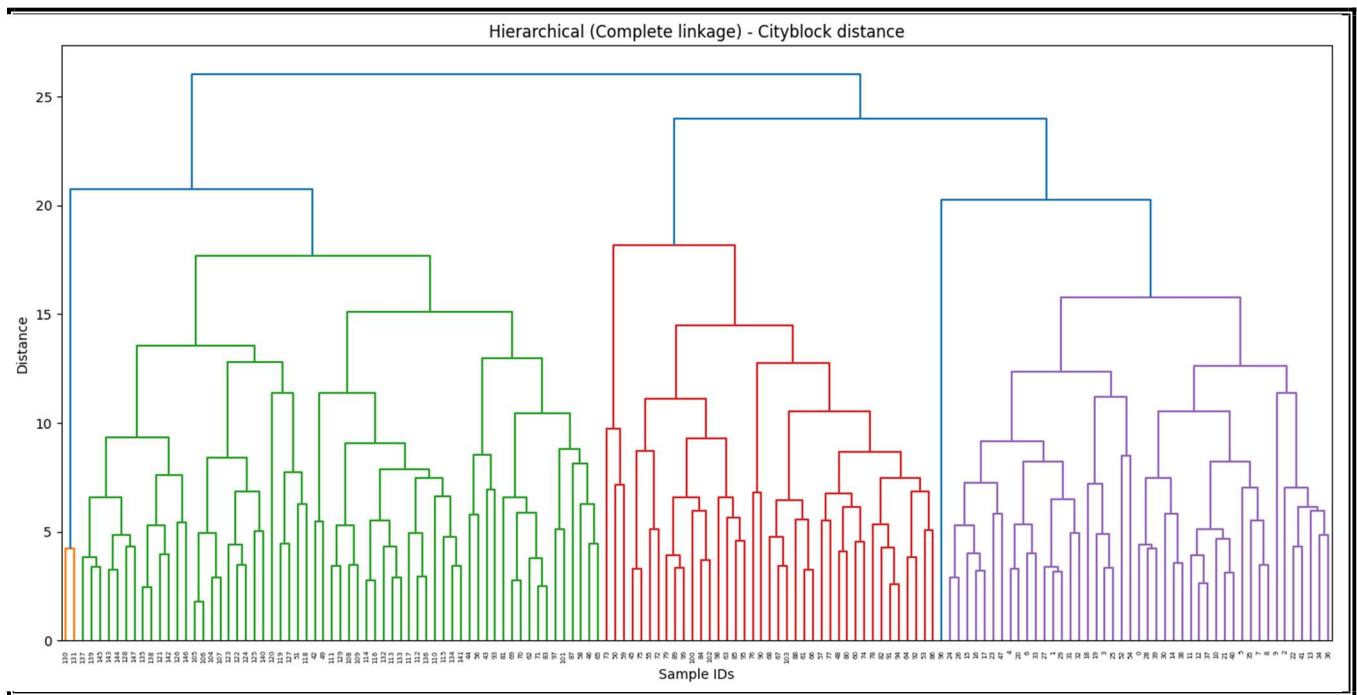


Figure 6: Dendrogram obtained on performing Hierarchical Clustering using Complete Linkage with Manhattan Distance as the distance metric and $t = 3$ maximum number of clusters

- The ARI and AMI values for each of the clustering strategies described above are given as follows:

Type of Linkage	Distance Metric	ARI	AMI
Single	Euclidean	-0.010913368924245862	0.013447552615202992
Single	Manhattan	-0.010913368924245862	0.013447552615202992
Complete	Euclidean	0.745740896797098	0.7620088581447081
Complete	Manhattan	0.5958407796009646	0.6716204925118936

Table 2: ARI and AMI value for different Hierarchical Clustering approaches based on linkage and distance metrics

- The negative value of ARI for clustering approaches relying on Single Linkage reflect poor clustering. These clustering approaches also report low AMI values, indicating that the predicted and real clusters have a low extent of agreement.
- However, clustering approaches relying on Complete Linkage reflect a marked increase in the ARI and AMI values, regardless of the distance metric used.
- Among these, **clustering performed using Complete Linkage and Euclidean Distance** provides **the highest set of values for ARI and AMI**; moreover, these values are higher than those corresponding values obtained using Ward Linkage and Euclidean Distance in the previous example.
- Therefore, this choice of linkage measurement and distance metric is the best suited for accurately clustering the given dataset (as indicated by the high ARI value), such that the predicted clusters match their real counterparts to a substantial extent (as indicated by the high AMI value).

3.

- The variation of Silhouette Scores with the number of clusters 'k' is given as follows:

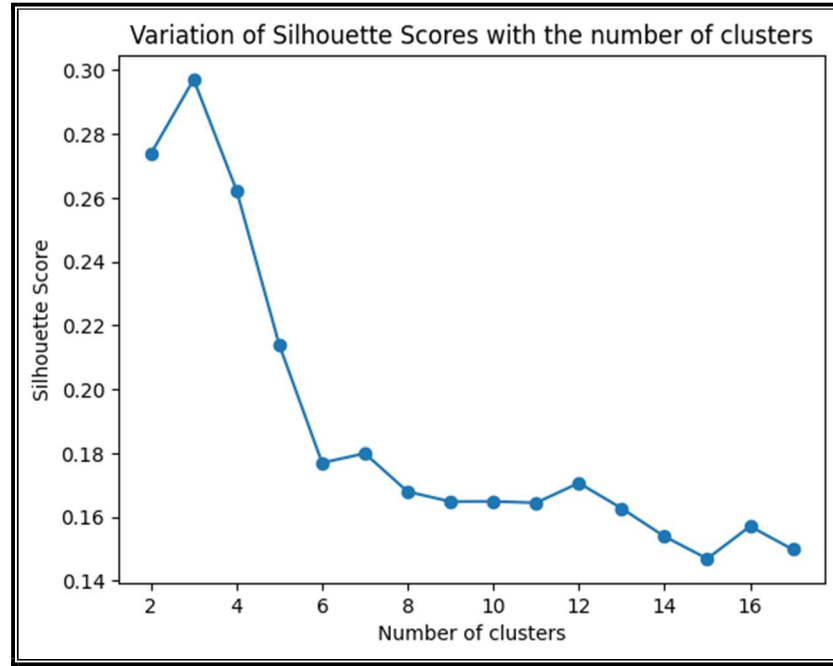


Figure 7: Variation of Silhouette Scores with the number of clusters 'k'

- The variation of CH Indices with the number of clusters 'k' is given as follows:

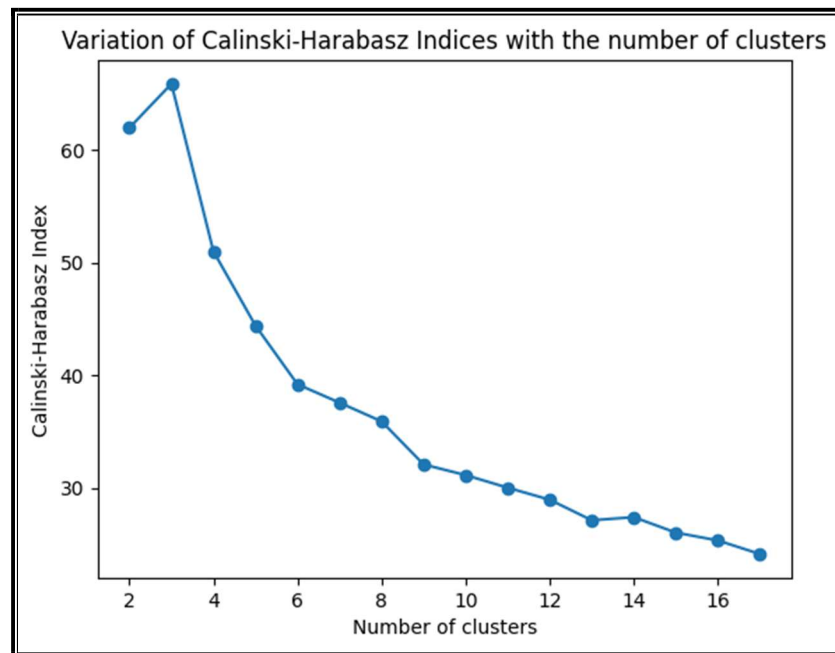


Figure 8: Variation of Calinski-Harabasz Indices with the number of clusters 'k'

- Relatively high and positive values of the (average) Silhouette Score, which ranges between -1 and 1, indicate that data points are well-matched to their correct clusters; alternatively, relatively few data-points are incorrectly clustered in the predicted clusters,
- Relatively high and positive values of the CH Index indicate that samples assigned to a cluster are more similar to other samples belonging to the same cluster than to the ones assigned to other clusters.

- Therefore, we determine the optimal 'k' value as the number of clusters where the Silhouette Score and CH Index plots achieve maximum values of the corresponding metric.
- Here, since each of the two plots above achieve their respective maxima at $k = 3$, we consider this value as the optimal 'k' value for further analyses.

4.

- K-Means Clustering was performed by setting $k = 3$, based on previous analyses. The ARI and AMI values obtained are given as follows:

ARI	0.7985910803225054
AMI	0.7926245997234933

Table 3: ARI and AMI values for K-Means Clustering by setting $k = 3$

- Thus, the performance of K-Means Clustering with $k = 3$ is comparable to that of Hierarchical Clustering performed using Complete Linkage and Euclidean Distance; in fact, the ARI and AMI values reported by K-Means Clustering are slightly higher than those reported by the other approach (from Tables 2 and 3).
- However, since both ARI and AMI are less than 1, the clustering is not fully accurate and some samples remain erroneously clustered and the predicted clusters are not identical to the real ones. This can be observed using the `kmeans_labels` dictionary provided in the code as well.
- For instance, sample with sample ID 44 has been clustered as belonging to Class 2, when it, in fact, belongs to Class 1.

5.

- DBSCAN was performed using $\epsilon = 2.2$ and $\text{min_samples} = 12$ (since we have 11 feature variables here). 35 outliers (out of the total 148) were identified and removed before performing further analyses; thus, subsequent analyses were performed using the remaining 113 samples.
- The ARI and AMI values obtained before and after removing outliers, using DBSCAN, are given as follows:

	Including outliers	Excluding outliers
ARI	0.6313720469125441	0.9738177626867184
AMI	0.6923028290793902	0.962220473133447

Table 4: ARI and AMI values including and excluding outliers, using DBSCAN

- We observe that the performance of DBSCAN, when outliers were included, was worse than that of Hierarchical Clustering with Complete Linkage and Euclidean Distance, and that of K-Means Clustering with $k = 3$.
- However, the performance of DBSCAN improved substantially when the outliers were removed; the values of both ARI and AMI were reported to be greater than 0.9, indicating that a large majority of the samples were correctly clustered and that the predicted clusters were in close agreement with the real ones.
- Hierarchical Clustering with Ward Linkage and Euclidean Distance was subsequently performed to compare the results obtained after removing outliers with those

obtained in Question 1. The 'maxclust' criterion was used to ensure that no more than $t = 3$ clusters were formed, while generating the clustering heatmap.

- The dendrogram obtained is given as follows:

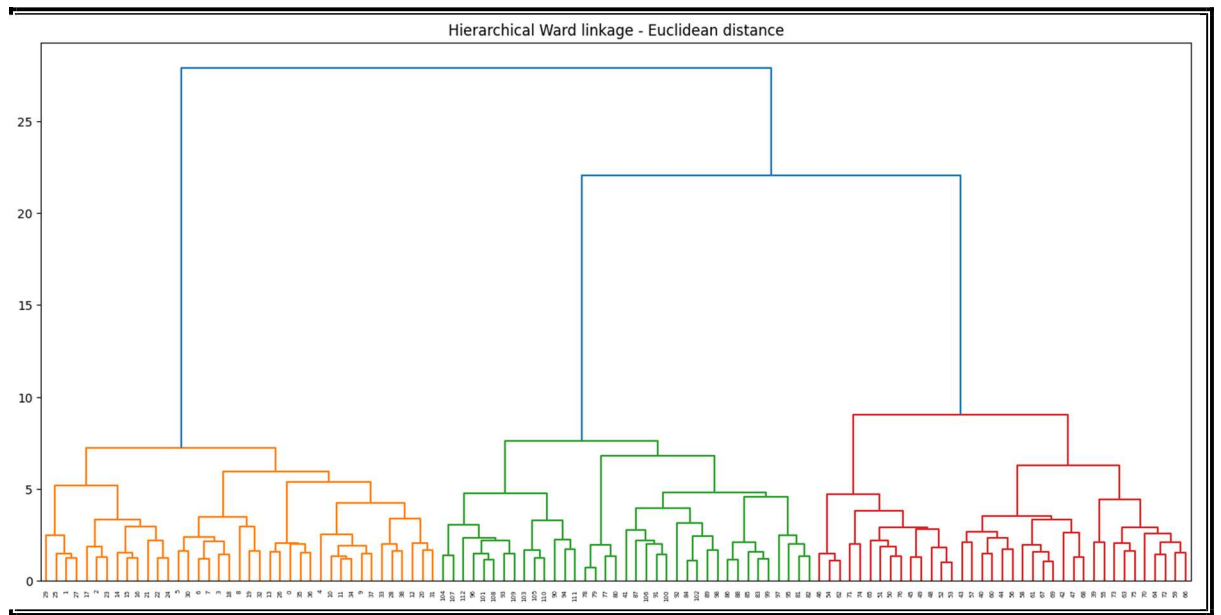


Figure 9: Dendrogram obtained on performing Hierarchical Clustering using Ward Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters, upon excluding outliers

- The ARI and AMI values obtained are given as follows:

ARI	0.9738177626867184
AMI	0.962220473133447

Table 5: ARI and AMI values for Hierarchical Clustering using Ward Linkage, upon excluding outliers

- The ARI and AMI values for Hierarchical Clustering with Ward Linkage substantially improved after removing the outliers. Thus, only a small minority of samples were erroneously clustered, which could be visualised using the following clustering heatmap with dendrogram:

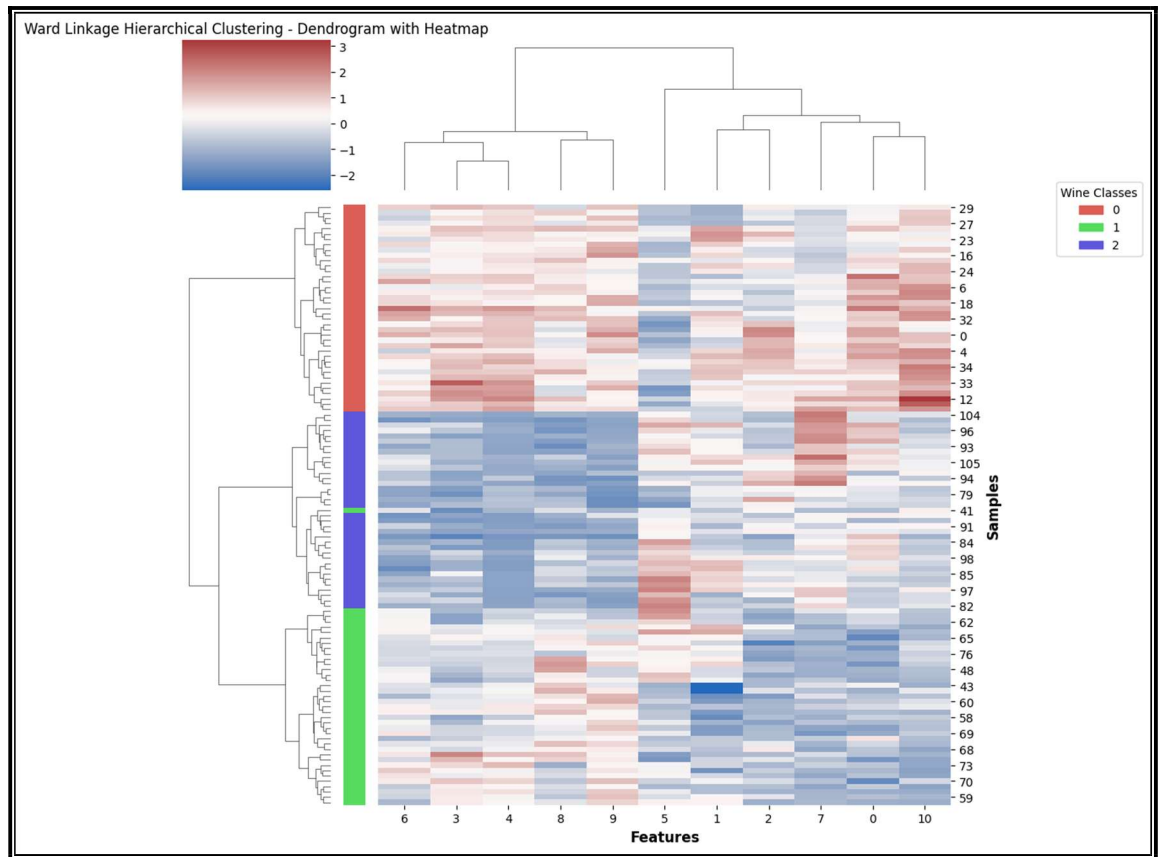


Figure 10: Heatmap with Dendrogram obtained on performing using Ward Linkage with Euclidean Distance as the distance metric and $t = 3$ maximum number of clusters, upon excluding outliers

- We observe that a relatively small number of samples belonging to Class 1 were erroneously clustered along with Class 2 samples; this is indicated by the presence of a narrow green band flanked by two wide blue bands in the above figure. Apart from these samples, all the remaining ones were clustered along with other samples belonging to the same class.
- Thus, the accuracy of clustering and the extent to which predicted clusters match with the real clusters improved substantially after excluding the outliers. This indicates that the erroneously clustered samples observed in the previous implementation of Hierarchical Clustering were, in fact, the outliers.
- Furthermore, K-Means Clustering was implemented by setting $k = 3$, for the sake of consistency and comparison with the clustering performed in Question 4.
- The ARI and AMI values obtained are given as follows:

ARI	0.9483592276189997
AMI	0.9357927133205064

Table 6: ARI and AMI values for K-Means Clustering by setting $k = 3$, upon excluding outliers

- Thus, the values of ARI and AMI improved markedly upon removing outliers as compared to those obtained in Question 4. Thus, the accuracy of clustering and the extent to which predicted clusters match with the real clusters improved substantially after excluding the outliers.
- Thus, removing outliers substantially improved the performance of all the three clustering methods considered here, including Hierarchical Clustering, K-Means Clustering and DBSCAN.
- However, we note that the ARI and AMI values obtained by performing K-Means Clustering, on removing outliers, were marginally lower than those obtained by performing Hierarchical Clustering using Ward Linkage and Euclidean Distance, upon excluding outliers (from Tables 5 and 6).
- Therefore, removing outliers and performing Hierarchical Clustering using Ward Linkage and Euclidean Distance produced the best performance among the various clustering methods considered here.

References:

1. <https://www.geeksforgeeks.org/machine-learning/clustering-metrics/>